

Due: Thursday March 10th, latest at 2PM

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This problem set is about magnetic fields and the trajectories of charged particles in magnetic fields.

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1. Maxwell's 2nd equation asserts that the magnetic field has zero divergence.

(a) [3 pts] The vector field

$$\mathbf{V}_1 = x\hat{i} + y^2\hat{j} + 3\hat{k}$$

is claimed to describe the magnetic field in some region of space. Show that this would violate Maxwell's second equation.

(b) [3 pts] The vector field $\mathbf{V}_2 = \frac{y\hat{i} - x\hat{j}}{x^2 + y^2}$ is claimed to describe a magnetic field in some region of space. Is this claim compatible with Maxwell's second equation?

2. Steady current I flows through an infinitely long straight wire placed along the z axis, from $z = -\infty$ to $z = +\infty$ through the origin.

(a) [2 pts] Sketch a top view of the x - y plane, and show the point P with coordinates $(1, 4)$ and the point Q with coordinates $(3, 2)$. For each of these points, draw the direction of the magnetic field produced by the steady current I through the infinite wire.

(b) [4 pts] Calculate the magnitudes of the magnetic field created at those two points (P and Q).

(c) [4 pts] What is the magnitude of the magnetic field at a generic point (x, y) on the x - y plane? (A sketch similar to the previous one might help, in order to find the necessary distance.)

(d) [4 pts] At a generic point (x, y) on the x - y plane, find the x -, y -, and z - components of the magnetic field. (A sketch might help.)

(e) [4 pts] Now consider a completely generic point (x, y, z) , not necessarily on the x - y plane. What is the magnitude of the magnetic field at this point? What are the x -, y -, and z - components of the magnetic field at this point?

3. An electron (charge $\approx 1.602 \times 10^{-19}$ C; mass $\approx 9.109 \times 10^{-31}$ kg) moving at speed 100 m/s finds itself in a region of constant magnetic field. The magnetic field points perpendicular to the electron velocity and has strength 10^{-4} Tesla.

- (a) [4 pts] Find the magnitude of the magnetic force experienced by the electron.
- (b) [4 pts] The magnetic force causes cyclotron motion, so that the electron moves around a circular trajectory. Find the radius of the circular trajectory.

4. A particle with charge Q and mass m is placed in a region with constant electric field in the y -direction, $\mathbf{E} = E\hat{j}$, and constant magnetic field in the z direction, $\mathbf{B} = B\hat{k}$. The particle position coordinates are x, y, z , and its velocity components are denoted as v_x, v_y, v_z .

- (a) [4 pts] Calculate the force $\mathbf{F}$. Write out separately the expressions for its components, F_x, F_y, F_z . (These expressions will involve the quantities Q, B, E, v_x , and v_y .)
- (b) [4 pts] Using your results for F_x, F_y and Newton's 2nd law, $\mathbf{F} = m\frac{d\mathbf{v}}{dt}$, calculate expressions for $\frac{dv_x}{dt}$ and $\frac{dv_y}{dt}$. You should thus obtain coupled first-order differential equations for $v_x(t)$ and $v_y(t)$.
- (c) [4 pts] If the particle starts at rest, i.e., if the initial conditions are $v_x(0) = v_y(0) = 0$, then the solutions to the above equations are of the form

$$v_x = \frac{E}{B} - \frac{E}{B} \cos(\omega t) \quad v_y = \frac{E}{B} \sin(\omega t) \quad (1)$$

Substituting these into your first differential equation, find an expression for ω . Substituting these into your second differential equation, show that you find the same expression for ω .

- (d) [4 pts] The equations (1) are themselves differential equations for $x(t)$ and $y(t)$, since $v_x = \frac{dx}{dt}$ and $v_y = \frac{dy}{dt}$. Assume that the particle starts at the origin, so that the initial conditions are $x(0) = y(0) = 0$. Find the solutions for $x(t)$ and $y(t)$.
- (e) [2 pts] Sketch $x(t)$ as a function of t and $y(t)$ as a function of t . Sketch the trajectory of the particle in the x - y plane.